Topic 1: What is Generative AI?

1.1 Definition and Core Idea

- Generative AI refers to artificial intelligence systems that can create new content -- text, images, audio, video, and code -- rather than just analysing or classifying existing content
- Traditional (discriminative) AI: takes input, produces a label or decision (spam/not spam, cat/dog)
- Generative AI: takes input (a prompt, an image, a melody) and produces entirely new content
- The content it generates did not exist before -- it is synthesised from learned statistical patterns
- Examples: ChatGPT writes essays, DALL-E 3 creates images, Suno composes music, GitHub Copilot writes code

1.2 How is it Different from Traditional AI?

- Discriminative models learn the boundary between categories: 'Is this email spam?'
- Generative models learn the underlying data distribution: 'What does a legitimate email look like?'
- Traditional AI output: a class label, a probability, a numeric prediction
- Generative AI output: text, images, audio, video, code, 3D models, synthetic data
- Both are machine learning -- generative AI is simply applied to the task of creation

1.3 Why Did Generative AI Emerge Now?

- Transformers (2017): a new neural network architecture that processes entire sequences in parallel, enabling massive scale
- Scale: models trained on hundreds of billions of parameters and trillions of text tokens unlocked emergent abilities
- Data: the internet provided unprecedented volumes of human-generated text, images, and code for training
- Hardware: GPU and TPU improvements made training and inference economically viable at large scale
- The result: models that can hold conversations, write software, pass professional exams, and generate photorealistic images

1.4 Key Terms to Know

- Prompt: the input you give the model -- a question, instruction, or example
- Token: the unit of text a model processes (roughly one word or sub-word); pricing and context limits are measured in tokens
- Context window: the maximum amount of text (in tokens) the model can consider at once
- Inference: running the model to generate output (as opposed to training)

- Hallucination: when a model generates confident but factually incorrect information